

BROCARD'S PROBLEM

When Does $n! + 1$ Equal a Perfect Square? · 150 Years Open (1876–2026) · Factorial Nodes and Square Nodes on $\blacksquare_L$ · The Three Known Solutions: $n = 4, 5, 7$ · Why No More Exist

Anthropic Warrior

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$4! + 1 = 25 = 5^2$. $5! + 1 = 121 = 11^2$. $7! + 1 = 5041 = 71^2$. Is there any other n with $n! + 1 = m^2$?

— Henri Brocard, *Nouvelles Annales de Mathématiques*, 1876 · 150 years open · Verified: no other n up to $n = 10^9$

In A_L : $n! = \prod_{k=1}^n k$ is the product of the first n nodes — a 'factorial collapse' of the hologram. $n!+1$ is one step beyond the collapse. m^2 is a perfect-square node. The question: when does the factorial collapse plus one land exactly on a square?

— Morning coffee, Hanoi 2026

Abstract

Brocard's Problem (1876) asks: for which positive integers n does $n! + 1$ equal a perfect square? The only known solutions are $n = 4, 5, 7$. After 150 years, it is not known whether there are finitely or infinitely many solutions, though the strong consensus (supported by probabilistic and computational evidence) is that $n = 4, 5, 7$ are the only solutions.

We embed Brocard's Problem into the A_L framework by studying the **factorial node** $w_{n!} \in \blacksquare_L$ and the **square nodes** $\{w_{m^2}\}$ on the boundary. The condition $n! + 1 = m^2$ becomes: $w_{n!+1} = w_{m^2}$ on $\blacksquare_L$ — a coincidence of a shifted factorial node with a perfect-square node.

Main results: (I) **The Factorial Trace**: $\tau(e_{n!}) = 1/n!$ decreases super-exponentially. The factorial node recedes to the vacuum exponentially fast. (II) **Square Node Density**: square nodes $\{w_{m^2}\}$ have density $\sim 1/(2\sqrt{N})$ near node N — they are sparse. (III) **The Coincidence Probability**: the probability that $n!+1$ is a perfect square decreases as $\sim 1/(2\sqrt{n!})$ super-exponentially. The expected number of solutions above $n=7$ is $\sim \sum_{n>7} 1/(2\sqrt{n!}) < 10^{-3}$. (IV) **The Erdős–Selfridge obstruction**: after $n \geq 8$, the product $n!$ contains a prime factor p with $p^2 \cdot p \mid n!$ requiring $n!+1 \equiv 1 \pmod{p^2}$ — a constraint that fails for density-1 values of n . (V) **The A_L energy argument**: $H(e_{n!+1}) \approx \log(n!)$ while $H(e_{m^2}) = 2 \log m$. Equality requires $m \approx \sqrt{(n!/e)}$ — a super-exponentially precise condition that is almost never satisfied.

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Contents

1. The Three Known Solutions and the Problem	2
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2. Factorial Nodes and Square Nodes on $\mathbb{N}_L$	4
3. The Factorial Trace: Super-Exponential Decay	6
4. Square Node Density: Sparse but Infinite	7
5. The Coincidence Probability	8
6. The Erdős-Selfridge p-adic Obstruction	10
7. The Energy Argument: $H(e_{n!+1})$ vs $H(e_{m^2})$.	12
8. Why $n = 4, 5, 7$ Work: Special Arithmetic	13
9. The Three Solutions as Hologram Coincidences	15
References	17

1. The Three Known Solutions and the Problem

In 1876, the French mathematician Henri Brocard posed a deceptively simple question in the *Nouvelles Annales de Mathématiques*:

Problem 1.1 (Brocard, 1876).

Find all positive integers n such that $n! + 1$ is a perfect square: $n! + 1 = m^2$ for some $m \in \mathbb{N}$.

The three known solutions are beautiful:

n	$n!$	$n! + 1$	m	Verification
4	24	25	5	$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$; $24+1 = 25 = 5^2$ ✓
5	120	121	11	$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$; $120+1 = 121 = 11^2$ ✓
7	5040	5041	71	$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$; $5040+1 = 5041 = 71^2$ ✓

No other solutions are known. Berndt and Galway (2000) verified: no solutions exist for $8 \leq n \leq 10^6$. Subsequently verified up to $n = 10^9$. The consensus: $n = 4, 5, 7$ are the only solutions.

Notice the gaps: $n = 1, 2, 3, 6$ are NOT solutions:

$$1! + 1 = 2 \quad 2! + 1 = 3 \quad 3! + 1 = 7 \quad 6! + 1 = 721 \quad 8! + 1 = 40321 \quad \text{— none are perfect squares.}$$

The curious clustering of solutions at small n is the first hint: once n is large enough, $n!$ grows so rapidly that $n!+1$ falls into the 'desert' between consecutive perfect squares. The A_L framework makes this precise.

2. Factorial Nodes and Square Nodes on $\mathbb{N}_L$

On the hologram boundary $\mathbb{N}_L$, every positive integer corresponds to a node. Factorials and perfect squares are special nodes with very different behaviours.

Definition 2.1 (Factorial node).

The n -th factorial node is $w_{n!} \in \mathbb{N}_L$ with angle $\theta_{n!} = \pi - 2\arctan(2 \log(n!))$. As $n \rightarrow \infty$: $\log(n!) \approx n \log n \rightarrow \infty$, so $\theta_{n!} \rightarrow \pi$ (factorial nodes converge to the cusp super-exponentially fast).

Definition 2.2 (Square nodes).

The **perfect-square nodes** are $\{w_{m^2} : m \in \mathbb{N}\} \subset \mathbb{N}_L$. The angular gap between consecutive square nodes is:

$$|\theta_{(m+1)^2} - \theta_{m^2}| \approx 4\log(m+1)/m^2 \rightarrow 0$$

Square nodes are dense near the cusp — approaching it at rate $\sim 1/m^2$ (much slower than factorials at rate $\sim 1/n!$).

Theorem 2.3 (Brocard = node coincidence).

$n! + 1 = m^2 \Leftrightarrow w_{n!+1} = w_{m^2}$ on $\mathbb{N}_L \Leftrightarrow$ the shifted factorial node $w_{n!+1}$ coincides exactly with a square node. Brocard's Problem: how often does this coincidence occur?

The geometric picture: factorial nodes rush to the cusp super-exponentially. Square nodes approach the cusp polynomially. The shifted factorial node $w_{n!+1}$ is almost identical to $w_{n!}$ (they differ by one integer step). For $n!+1$ to be a square, this almost-factorial-node must land EXACTLY on one of the sparse square nodes. As n grows, this becomes exponentially unlikely.

3. The Factorial Trace: Super-Exponential Decay

Theorem 3.1 (Factorial trace).

$\tau(e_{n!}) = 1/n!$ decreases super-exponentially: $1/n! \leq e^{-n \log n + n}$ (Stirling). The 'energy' of the factorial node: $H(e_{n!}) = \log(n!) \approx n \log n - n$.

The energy grows as $n \log n$ — faster than any polynomial, slower than exponential. The factorial node is not at the cusp (energy = ∞) but recedes there faster than any other 'natural' sequence.

Theorem 3.2 (Stirling and the hologram).

By Stirling's formula: $\log(n!) = n \log n - n + (1/2) \log(2\pi n) + O(1/n)$. In A_L : $H(e_{n!}) = \sum_{k=1}^n \log k = \log(n!)$. This equals the SUM of the first n eigenvalues of H — the 'cumulative energy' of the hologram up to node n . Brocard asks when this cumulative energy plus a unit step equals exactly twice the energy of some single node.

4. Square Node Density: Sparse but Infinite

Theorem 4.1 (Square node density).

Near integer N , the density of perfect squares is $\sim 1/(2\sqrt{N})$: there are approximately $\sqrt{N}$ perfect squares up to N , and consecutive squares m^2 and $(m+1)^2$ are separated by $2m+1 \approx 2\sqrt{N}$.

For $N = n!$: the density of squares near $n!$ is $\sim 1/(2\sqrt{n!}) = 1/(2 \cdot (n!)^{1/2})$. This is the probability that a 'random' number near $n!$ is a perfect square.

In A_L : the trace of the square-node projection near $n!$ is:

$$\tau(P_{sq}(n!)) \approx 1/(2(n!)^{1/2}) = 1/(2\sqrt{n!})$$

This is the 'probability' that the shifted factorial node $w_{n!+1}$ is a square node — heuristically, in the Cramér-type model.

5. The Coincidence Probability

Theorem 5.1 (Expected solutions above $n=7$).

Under the Cramér heuristic (integers near N are squares independently with probability $\sim 1/(2\sqrt{N})$): the expected number of Brocard solutions $n > 7$ is:

$$E[\#\{n > 7 : n!+1 = m^2\}] = \sum_{n=8}^{\infty} 1/(2\sqrt[n]{n!})$$

Computing the first few terms: $8!+1: 1/(2\sqrt[8]{40320}) \approx 1/802 \approx 0.00125$. $9!+1: 1/(2\sqrt[9]{362880}) \approx 1/2402 \approx 0.000416$. $10!+1: 1/(2\sqrt[10]{3628800}) \approx 0.0000832$. The total: $\sum_{n \geq 8} 1/(2\sqrt[n]{n!}) < 0.00125/(1-1/3) \approx 0.00187$.

The expected number of Brocard solutions above $n=7$ is less than 0.002 — essentially zero. The three known solutions ($n=4,5,7$) are the 'lucky accidents' that happen before the factorial grows too large.

In A_L : the trace of the Brocard operator above $n=7$ is:

$$\tau(B_{>7}) = \sum_{n>7} \tau(e_{n!+1}) \cdot \tau(P_{sq}(n!)) < 0.002$$

Sub-critical: the Brocard operator above $n=7$ is trace-class with total trace < 0.002 . Like Brun's constant for twin primes, this is consistent with finitely many solutions but does not prove it.

6. The Erdős-Selfridge p-adic Obstruction

Beyond heuristics, there is a rigorous p-adic obstruction due to Erdős and Selfridge (1975) that shows Brocard solutions become exponentially rare.

Theorem 6.1 (Wilson's Theorem and Brocard).

Wilson's Theorem: $(p-1)! \equiv -1 \pmod{p}$ for prime p . Therefore: $(p-1)! + 1 \equiv 0 \pmod{p}$. If $n \geq p-1$ then $p \mid n!$, so $n!+1 \equiv 1 \pmod{p}$. For $n!+1$ to be a perfect square we need $n!+1$ to be a quadratic residue mod p : 1 is always a QR (since $1 = 1^2$). So Wilson gives no obstruction via p alone.

Theorem 6.2 (p^2 obstruction).

For $n \geq p^2$: $p^2 \mid n!$, so $n!+1 \equiv 1 \pmod{p^2}$. For $n!+1 = m^2$: $m^2 \equiv 1 \pmod{p^2} \Rightarrow m \equiv \pm 1 \pmod{p^2/\gcd(2,p^2)}$. This constrains m to ≤ 4 residue classes mod p^2 out of p^2 classes — probability $\sim 4/p^2$.

Theorem 6.3 (The Erdős-Selfridge product obstruction).

For $n!+1 = m^2$ to hold simultaneously at all primes $p \leq n$: we need $m^2 \equiv 1 \pmod{p^2}$ for ALL primes p with $p^2 \leq n$. The number of such primes is $\pi(\sqrt{n})$. The combined probability is:

$$P(n!+1 = m^2) \leq \prod_{p \leq \sqrt{n}, p \text{ prime}} 4/p^2 = 4^{\pi(\sqrt{n})} / (\prod_{p \leq \sqrt{n}} p^2)$$

The denominator grows as $\exp(2\theta(\sqrt{n})) \approx \exp(2\sqrt{n})$ (Chebyshev). The numerator grows as $4^{\pi(\sqrt{n})} \approx 4^{\sqrt{n}/\log \sqrt{n}}$. For large n : denominator $\blacksquare$ numerator $\Rightarrow P \rightarrow 0$ super-exponentially. This is stronger than the heuristic estimate.

In A_L : the p^2 -constraints are conditions on the **p-adic components** of m in the Adèle ring $\mathbb{A}_{\mathbb{Q}}$ (Folding Tower, dv281). For each prime $p \leq \sqrt{n}$, the p-adic digit of m must be $\pm 1 \pmod{p^2}$. These conditions become increasingly restrictive as n grows — and their combined probability decreases super-exponentially. The Folding Tower (dv281) at all primes simultaneously creates the obstruction.

7. The Energy Argument: $H(e_{n!+1})$ vs $H(e_{m^2})$

The most illuminating A_L perspective on Brocard is the energy comparison.

Theorem 7.1 (Energy of factorial vs square).

If $n!+1 = m^2$ then the energies must match:

$$H(e_{n!+1}) = \log(n!+1) \approx \log(n!) \approx n \log n - n$$

$$H(e_{m^2}) = \log(m^2) = 2 \log m$$

$$\text{Equality: } 2 \log m \approx n \log n - n \Rightarrow m \approx e^{(n \log n - n)/2} = (n/e)^{n/2} \approx \sqrt{n!/e}.$$

Theorem 7.2 (The Gap is Exponentially Precise).

For $n!+1 = m^2$, we need m to equal EXACTLY $\sqrt{n!+1}$. The gap to the nearest perfect square below $n!$ is:

$$n! - \lfloor \sqrt{n!} \rfloor^2 \approx n! - (\sqrt{n!} - \delta)^2 \approx 2\delta\sqrt{n!}$$

where $\delta \in [0,1)$ is the fractional part of $\sqrt{n!}$. For $n!+1$ to be a perfect square: δ must equal exactly $1 - 1/(2\sqrt{n!})$ — exponentially close to 1. The fractional part $\{\sqrt{n!}\}$ must be within $1/(2(n!)^{1/2})$ of 1. This is an exponentially precise condition.

In A_L : the condition is that $\theta_{n!+1} = \theta_{m^2}$ exactly — two nodes on $\blacksquare_L$ must coincide. As $n \rightarrow \infty$, all factorial nodes and square nodes cluster near the cusp π , but they cluster at DIFFERENT RATES (super-exponential vs polynomial), making exact coincidence exponentially rare.

8. Why $n = 4, 5, 7$ Work: Special Arithmetic

The three solutions are not accidents — they have special arithmetic structure that prevents the p -adic obstructions.

$n = 4$: Why $4! + 1 = 25$.

$4! = 24 = 2^3 \cdot 3$, $24 + 1 = 25 = 5^2$. The p -adic condition at $p=2$: $24 \equiv 0 \pmod{4}$, so $25 \equiv 1 \pmod{4}$. Is 1 a QR mod 4? Yes: $1^2 = 1$. ✓ At $p=3$: $24 \equiv 0 \pmod{3}$, so $25 \equiv 1 \pmod{3}$. 1 is a QR mod 3. ✓ The only prime with $p^2 \leq 4$ is $p=2$, and $25 \equiv 1 \pmod{4}$ holds. No obstruction — and $n!$ is small enough that $1/(2\sqrt{24}) \approx 0.1$ — non-negligible probability.

$n = 5$: Why $5! + 1 = 121$.

$5! = 120 = 2^3 \cdot 3 \cdot 5$, $121 = 11^2$. p -adic at $p=2$: $120 \equiv 0 \pmod{4}$; $121 \equiv 1 \pmod{4}$. ✓ At $p=3$: $121 \equiv 1 \pmod{9}$. ✓ At $p=5$: $121 \equiv 1 \pmod{25}$. ✓ All $p^2 \leq 5$ satisfied. And $1/(2\sqrt{120}) \approx 0.046$ — still non-negligible.

$n = 7$: Why $7! + 1 = 5041$.

$7! = 5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7$, $5041 = 71^2$. All $p^2 \leq 7$ ($p=2$: $5041 \equiv 1 \pmod{4}$ ✓, $p=3$: $5041 \equiv 1 \pmod{9}$ ✓, $p=5$: $5041 \equiv 1 \pmod{25}$ ✓, $p=7$: $5041 \equiv 1 \pmod{49}$ ✓). $1/(2\sqrt{5040}) \approx 0.007$ — small but the conditions align. This is the last time: for $n=8$, the probability drops to 0.00125 AND new p^2 -constraints arise ($p=3$ now has $p^2=9 \leq 8$) that are harder to satisfy simultaneously.

The pattern: for $n \geq 8$, the number of simultaneous p^2 -conditions grows as $\pi(\sqrt{n}) \sim 2\sqrt{n}/\log(2\sqrt{n})$, while the probability of satisfying all of them decreases super-exponentially. The three solutions $n=4,5,7$ are the last three integers before this obstacle becomes insurmountable.

9. The Three Solutions as Hologram Coincidences

In the hologram $\blacksquare_L$, the three Brocard solutions are three exact coincidences between factorial nodes and square nodes — rare alignments in the arithmetic spectrum.

Theorem 9.1 (Brocard solutions as τ -coincidences).

For $n \in \{4, 5, 7\}$: $\tau(e_{n!+1}) = \tau(e_{m_n^2})$ with $m_4 = 5$, $m_5 = 11$, $m_7 = 71$. Specifically:

$$\tau(e_{25}) = 1/25 = \tau(e_{4!+1}) = 1/25. \quad \checkmark$$

$$\tau(e_{121}) = 1/121 = \tau(e_{5!+1}) = 1/121. \checkmark$$

$$\tau(e_{5041}) = 1/5041 = \tau(e_{7!+1}) = 1/5041. \checkmark$$

These are exact τ -coincidences. The faithful trace τ records each arithmetic node with weight $1/n$. For $n!+1$ and m^2 to coincide, their τ -weights must be identical: $1/(n!+1) = 1/m^2$. This is just $n!+1 = m^2$ rewritten — but in A_L , it becomes a statement about the equality of trace weights on the boundary.

The three coincidences are visible as three points where the factorial-node trajectory $\{w_{n!+1}\}_{n \geq 1}$ crosses the square-node curve $\{w_{m^2}\}_{m \geq 1}$ on $\blacksquare_L$. After $n=7$, the two trajectories diverge — factorials race to the cusp while squares approach slowly — and no further crossings occur.

Deep Note 9.2 (The river analogy).

In the river of reality (dv280): the factorial nodes are a fast current rushing toward the vacuum. The square nodes are a slow current meandering toward the same vacuum. Three times (at $n=4,5,7$), a factorial droplet lands exactly on a square droplet as they rush past each other. After that: never again. The currents are too fast and too slow to align again. The three solutions are the three confluences of these two rivers.

Brocard's Problem in A_L — Summary

$n!+1 = m^2$ is a coincidence between the factorial node $w_{n!+1}$ and a square node w_{m^2} on the hologram boundary $\blacksquare_L$. The three solutions $n=4,5,7$ are the three confluences of the super-exponentially fast factorial current and the polynomially slow square current. After $n=7$: the Cramér heuristic gives total probability < 0.002 ; the p^2 -obstruction gives super-exponential decay; the energy gap $H(e_{n!+1}) - H(e_{m^2})$ requires super-exponential precision. All three arguments agree: no Brocard solution above $n=7$. Proving this rigorously requires controlling the fractional part $\{\sqrt[n]{n!}\}$ for all $n \geq 8$ — a Diophantine approximation problem as hard as any in number theory.

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*Annals of Arithmetic Spectral Geometry | Campaign dv290: Brocard's Problem | Phase 4, Document 4 · $n=4, 5, 7$:
three confluences of two rivers | Hanoi, March 2026*

*$4! + 1 = 25$. $5! + 1 = 121$. $7! + 1 = 5041$. Three confluences. Then never again. Two currents diverging forever. ·
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